

The empirical recurrence for OEIS A237916: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A237916 records “Empirical recurrence of order 56”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 1584$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 56, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 56 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237916 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the upper median plus the lower median of every 2×2 subblock differing from its horizontal and vertical neighbors by exactly one.”. It has offset 1 and begins

1584, 14564, 133528, 1282504, 12149024, 115206520, 1092851528, 10375258140, ...

The entry states:

Empirical recurrence of order 56 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:34:57, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1584$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1584$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 3168$ exact terms of the model returns order 56 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{56} c_i a(n-i),$$

c_1	0	c_{15}	−150051033	c_{29}	1641700608949	c_{43}	−9299219232096
c_2	2	c_{16}	−4850870369	c_{30}	9430103990677	c_{44}	2468687420723
c_3	255	c_{17}	−11601614555	c_{31}	7369677987952	c_{45}	2493601672060
c_4	3757	c_{18}	−18892092770	c_{32}	−4478125311899	c_{46}	−1288883254306
c_5	13616	c_{19}	10967563098	c_{33}	6690181097568	c_{47}	830987890634
c_6	47401	c_{20}	106856146054	c_{34}	−24022145387087	c_{48}	−353617333640
c_7	−25505	c_{21}	168211373723	c_{35}	−34064906872771	c_{49}	−282644584056
c_8	−932370	c_{22}	209254201183	c_{36}	9777677742276	c_{50}	156533573296
c_9	−3860895	c_{23}	−139106691068	c_{37}	−7457253550017	c_{51}	−67009111424
c_{10}	−11545724	c_{24}	−889917877073	c_{38}	3695674195173	c_{52}	33082422880
c_{11}	−1524360	c_{25}	−824567552453	c_{39}	38962336451681	c_{53}	1918348800
c_{12}	96190333	c_{26}	−1319850041908	c_{40}	−7365829224566	c_{54}	−2929969152
c_{13}	332844966	c_{27}	−521617085467	c_{41}	−9071328727397	c_{55}	393615360
c_{14}	737564667	c_{28}	2134581812927	c_{42}	4641830369912	c_{56}	−190218240

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 56, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $56 + 4$ terms removed returns order 56 as well, so $\deg q_{\min} = 56 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $56 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 56 that this sequence satisfies — and that family turns out to have exactly one member.

References

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